

Chapter 9

Introduction

The ubiquitous use of computers and technology **prompts** some very important questions about the use of personal data and our right to privacy.

To efficiently and effectively use computers, you need to be aware of the potential impact of technology on people and how to protect yourself on the web.

People

Technology has had a very positive impact on people, but some of the impact could be negative.

- Most significant concerns:
 - Privacy – What are the threats to personal privacy and how can we protect ourselves?
 - Security – How can access to sensitive information be controlled and how can we secure hardware and software?
 - Ethics – How do the actions of individual users and companies affect society?

Privacy

- Privacy – **concerns** the collection and use of data about individuals
- Accuracy – responsibility of those who collect data
 - Must be secure and correct
- Property – relates to who owns the data
- Access – responsibility of those who control data and use that data

Large Databases

Large organizations compile information about us daily

- **Big Data** is exploding and ever-growing
 - 90% of the data collected has been collected over the last 2 years
- Information Resellers/Brokers
 - Collect and sell personal data
 - Create electronic profiles



Large Databases (Cont.)

Personal information is a marketable commodity, which raises many issues:

- Collecting public, **but personally identifying information** (e.g., Google's Street View)
- Spreading information without personal consent, leading to identity theft
- Spreading inaccurate information
 - Mistaken identity
- Freedom of Information Act
 - Entitlement to look at your records held by government agencies

Private Networks

Employee monitoring software

- Employers can monitor e-mail legally
 - A proposed law could prohibit this type of electronic monitoring or at least require the employer to notify the employee first

The Internet and the Web

- Illusion of anonymity
 - People are not concerned about privacy when surfing the Internet or when sending e-mail
- When browsing the web, critical information is stored on the hard drive in these locations:
 - History Files
 - Temporary Internet Files
 - Browser cache
 - Cookies
 - Privacy Mode
 - Spyware

History Files and Temporary Internet Files

History Files

- Include locations or addresses of sites you have recently visited

Temporary Internet Files / Browser Cache

- Saved files from visited websites
 - Offers quick re-display when you return to the site
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Cookies

Small data files that are deposited on your hard disk from web sites you have visited

- First-party cookies - generated only by websites you are visiting
- Third-party cookies - generated by an advertising company that is affiliated with the website
 - Also known as tracking cookies that keep track of your Internet activities through 3rd party cookies

Privacy Modes

- Ensures your browsing activity is not recorded on your hard drive
 - Incognito Mode
 - Google Chrome
 - Private Browsing
 - Safari

Privacy Threats

- Web bugs
 - Invisible images or HTML code hidden within an e-mail message or web page
- Spyware
 - Record and report Internet activities
 - Change browser to manipulate what you view
- Computer monitoring software
 - Most invasive and dangerous
 - Keystroke Loggers
 - Record activities and keystrokes
- Anti-Spyware programs
 - Detect and remove privacy threats

Online Identity

The information that people voluntarily post about themselves online

- Archiving and search features of the Web make it available indefinitely
- Major Laws on Privacy
 - **Gramm-Leach-Bliley Act** protects personal financial information
 - **Health Insurance Portability and Accountability Act (HIPAA)** protects medical records
 - **Family Educational Rights and Privacy Act (FERPA)** resists disclosure of educational records

Security

Involves protecting individuals or organizations from theft and danger

- Hackers
 - Gain unauthorized access with malicious intent
 - Not all hackers are illegal

Cybercrime / Computer Crime

- Criminal offense that involves a computer and a network
 - Effects over 400 million people annually
 - Costs over \$400 billion each year

Forms of Computer Crime

Computer Crime	Description
Identity theft	Illegal assumption of a person's identity for economic gain
Internet scams	Scams over the Internet
Data manipulation	Unauthorized access of a computer network and copying files to or from the server
Ransomware	Malicious software that encrypts your computer's data and ransoms the password to the user
DoS, Denial of service	Attempts to slow down or stop a computer system or network by flooding a computer or network with requests for information and data
DDoS, Distributed denial of service	Coordinates several computers making repeated requests for service

Internet Scams

Scams using the Internet.

- Internet scams have created financial and legal problems for many thousands of people
- Majority are initiated by a mass mailing to unsuspecting individuals

Common Internet Scams

Type	Description
Advance fee loans	Guaranteed low-rate loans available to almost anyone. After applicant provides personal loan-related information, the loan is granted subject to payment of an "insurance fee."
Auction fraud	Merchandise is selected and payment is sent. Merchandise is never delivered.
Fake antivirus software	A website or e-mail warns you that you are at risk of being infected by a computer virus and you need to download and install the security software they recommend. Ironically, the security software is fake and will install malicious software on your computer.
Nigerian Scam	A classic e-mail scam. The recipient receives an e-mail from a wealthy foreigner in distress who needs your bank account information to safely store their wealth, and for your troubles you will receive a large amount of money. Of course, once the scammer has your bank account information, your accounts will be drained and they will disappear

Social Engineering

Practice of manipulating people to divulge private data.

Played a key role in:

- Identity theft
- Internet scams
- Data manipulation
- Phishing
 - Attempts to trick Internet users into thinking a fake but official-looking website or e-mail is legitimate

Malicious Programs - Malware

- Malicious Programs Or Malware
 - Designed by crackers, computer criminals, to damage or disrupt a computer system
 - Computer Fraud and Abuse Act makes spreading a virus a federal offense
 - 3 most common programs
 - Viruses – migrate through networks and attach to different programs
 - Worms – fills the computer with self-replicating information
 - Trojan horse – programs disguised as something else

Malicious Hardware

Criminals use hardware for crimes.

Most common are:

- Zombies
 - Computers infected by a virus, worm, or Trojan Horse
 - Botnet or Robot Network is a collection of Zombies
- Rogue Wi-Fi Hotspots
 - Imitating legitimate free Wi-Fi
 - Capture data coming through the Rogue Wi-Fi
- Infect USB Flash Drives
 - Left on purpose in hopes for people to pick up and use
 - Have malicious software contained on them

Measures to Protect Computer Security

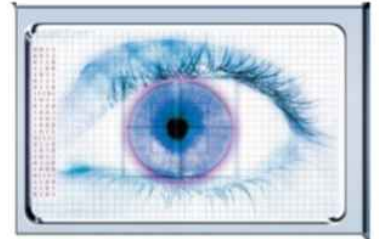
Principle measures to ensure computer security

- Computer Fraud and Abuse Act
 - Crime for unauthorized person to view, copy or damage data using computers across state lines
 - Prevents use of any government or federally insured financial institution computers

Measure	Description
Restricting access	Limit access to authorized persons using such measures as passwords, gestures, and biometric scanning.
Encrypting data	Code all messages sent over a network.
Anticipating disasters	Prepare for disasters by ensuring physical security and data security through a disaster recovery plan.
Preventing data loss	Routinely copy data and store it at a remote location.

Restricting Access

- Passwords
 - Dictionary attack
 - Uses software to try thousands of common words sequentially in an attempt to gain unauthorized access to a user's account
- Biometric scanning
 - Fingerprint scanners
 - Iris (eye) scanners
 - Facial recognition



Automated Security Tasks

Ways to perform and automate important security tasks

- Security Suites
 - Provide a collection of utility programs designed to protect your privacy and security
- Firewalls
 - Security buffer between a corporation's private network and all external networks
- Password Managers
 - Helps to create strong passwords

Encryption

Coding information to make it unreadable, except to those who have the encryption key or key

- The key will decrypt the information into a readable format

- Common uses for encryption:

- E-mail encryption
- File encryption
- Website encryption
 - HTTPS – hypertext transfer protocol secured
- Virtual private networks (VPNs)
- Wireless network encryption restricts access to authorized users
 - WPA2 – Wi-Fi Protected Access



Anticipating Disasters and Preventing Data Loss

Anticipating Disasters

- Physical Security protects hardware
- Data Security protects software and data from unauthorized tampering or damage
- Disaster Recovery Plan describes ways to continue operating in the event of a disaster

Preventing Data Loss

- Frequent backups
- Redundant data storage
 - Store off-site in case of loss of equipment

Making IT Work for You ~ Security and Technology

Precautions you as an individual can and should take to make sure that you aren't the victim of high-tech criminals

- Update software
- Be careful when browsing
- Be alert to e-mail scams
- Use antivirus software
- Strong passwords



Ethics

Standards of moral conduct

- Computer Ethics – guidelines for the morally acceptable use of computers
 - Copyright and Digital Rights Management
 - Cyberbullying
 - Plagiarism

Copyright and Digital Rights Management

- Copyright
 - Gives content creators the right to control the use and distribution of their work
 - Paintings, books, music, films, video games
- Software piracy
 - Unauthorized copying and distribution of software
 - Digital rights management (DRM) controls access to electronic media
 - Digital Millennium Copyright Act protects against piracy

Cyberbullying and Plagiarism

Cyberbullying

- Use of the Internet to send or post content intended to harm another person

Plagiarism

- Representing some other person's work and ideas as your own without giving credit to the original person's work and ideas



Careers in IT

- IT Security Analysts maintain the security of a company's network, systems, and data.
- Bachelors or associates degree in information systems or computer science
 - Experience is usually required
- Must safeguard information systems against external threats
- Annual salary is usually from \$58,000 to \$86,000
- Demand for this position is expected to grow



A Look to the Future ~ The End of Anonymity

- Most forums and comment areas on websites allow users to post messages anonymously
- Future software can identify you and track your moves through a mall or store

